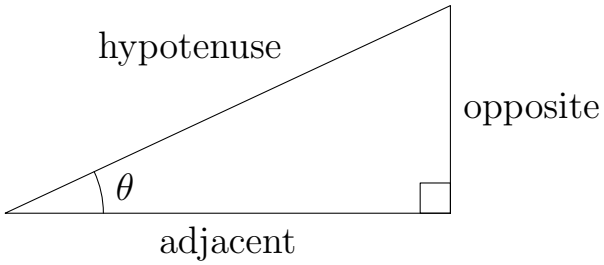
5.4: Right Angle Trigonometry

Definition.
Given a right triangle with an acute angle θ



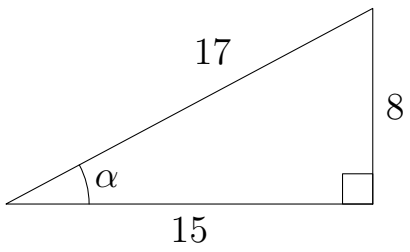
we have

$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$$

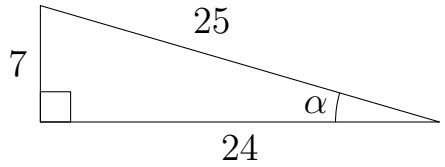
$$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$$

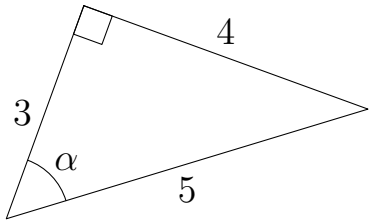
Example. Given the following triangle, find $\sin(\alpha)$, $\cos(\alpha)$, and $\tan(\alpha)$:



Example. Given the following triangle, find $\sin(\alpha)$:



Example. Given the following triangle, evaluate:

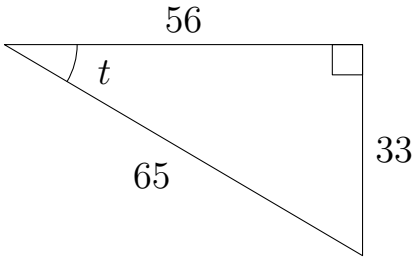


$\sin(\alpha)$ $\csc(\alpha)$

$\cos(\alpha)$ $\sec(\alpha)$

$\tan(\alpha)$ $\cot(\alpha)$

Example. Given the following triangle, evaluate:

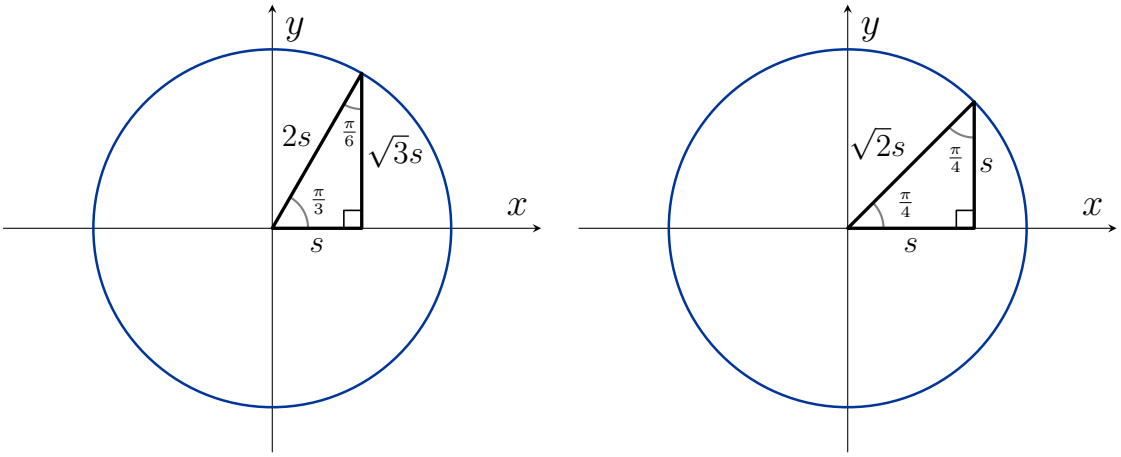


$\sin(t)$ $\csc(t)$

$\cos(t)$ $\sec(t)$

$\tan(t)$ $\cot(t)$

Side lengths of special angles:



Example. Find the exact value of the trigonometric functions of $\theta = \dfrac{\pi}{3}$, using side lengths.

$\sin(\theta)$ $\csc(\theta)$

$\cos(\theta)$ $\sec(\theta)$

$\tan(\theta)$ $\cot(\theta)$

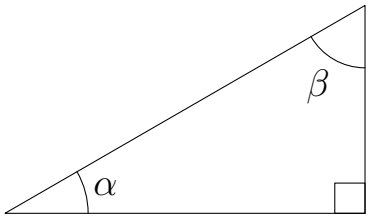
Example. Find the exact value of the trigonometric functions of $\vartheta = \dfrac{\pi}{4}$, using side lengths.

$\sin(\vartheta)$ $\csc(\vartheta)$

$\cos(\vartheta)$ $\sec(\vartheta)$

$\tan(\vartheta)$ $\cot(\vartheta)$

Equal cofunctions of complements:



$$\sin(\alpha) = \cos(\beta)$$
$$\sin(\beta) = \cos(\alpha)$$

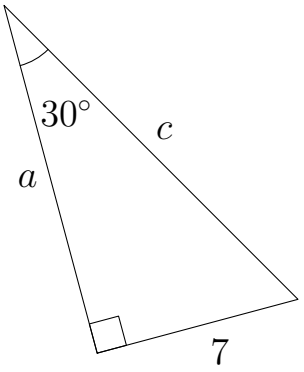
Definition. (Cofunction Identities)

$$\cos(\theta) = \sin\left(\frac{\pi}{2} - \theta\right) \qquad \sin(\theta) = \cos\left(\frac{\pi}{2} - \theta\right)$$
$$\sec(\theta) = \csc\left(\frac{\pi}{2} - \theta\right) \qquad \csc(\theta) = \sec\left(\frac{\pi}{2} - \theta\right)$$
$$\cot(\theta) = \tan\left(\frac{\pi}{2} - \theta\right) \qquad \tan(\theta) = \cot\left(\frac{\pi}{2} - \theta\right)$$

Example. If $\sin(t) = \frac{5}{12}$, find $\cos\left(\frac{\pi}{2} - t\right)$.

Example. If $\csc\left(\frac{\pi}{6}\right) = 2$, find $\sec\left(\frac{\pi}{3}\right)$.

Example. Find the unknown sides of the following triangle:



Example. To find the height of a tree, a person walks to a point 30 feet from the base of the tree. She measures an angle of 57° between a line of sight to the top of the tree and the ground. Find the height of the tree.

